

Implement a comprehensive monitoring system that combines acoustic sensing to detect lead dislodgement with thermal monitoring to detect tissue interface changes

Executive Technology Brief

Domain: acoustic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'adverse_event_without_identified_device_' for pacemaker lead: 16 MAUDE reports across 7 distinct devices (20220222 to 20230812); root-cause evidence from 1 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 1 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 4 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, injury, intervention, introducing, lead, maintain, matching, modes, must, problem, producing, recurring, report

Mechanism: A lifecycle-based risk management framework that integrates complication prevention across the entire continuum of pacemaker therapy

Architecture: PROPOSED architecture (no physical system exists): transducer element + matching layer; excitation / receive electronics; signal chain (detection criterion); coupling path to target medium — realizing: Implement a comprehensive monitoring system that combines acoustic sensing to detect lead dislodgement with thermal monitoring to detect tissue interface changes

Governing Model: Acoustics / ultrasonics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

AC-002: Wave speed relation. Expression: $c = f * \lambda$. Applicability to this invention: CONDITIONAL; holds under: non-dispersive propagation in the medium of interest. Assumptions: sound speed is medium- and temperature-dependent — source the value for the actual medium. Engaged variables: NOT ESTABLISHED.

AC-003: Exponential attenuation. Expression: $A = A_0 * \exp(-\alpha * d)$. Applicability to this invention: CONDITIONAL; holds under: homogeneous lossy medium; single frequency. Assumptions: α is frequency- and medium-dependent; two-way path doubles the exponent. Engaged variables: NOT ESTABLISHED.

AC-001: Acoustic impedance contrast (pulse-echo). Expression: $R = ((Z_2 - Z_1) / (Z_2 + Z_1))^2$. Applicability to this invention: CONDITIONAL; holds under: normal incidence, plane wave, lossless media. Assumptions: oblique incidence and attenuation modify detectability. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.